

Wrist and Hand Injuries in the Adolescent Athlete

Sonia Chaudhry

The incidence of hand injuries increases 20-fold after age 10, with sports being the most common mechanism of injury.¹ High impact sports such as football tend to produce injuries from single traumatic events, whereas repetitive overuse injuries are more common in sports such as gymnastics, golf, and tennis. We have focused on conditions and treatment considerations unique to the adolescent population.

One of the most challenging conditions is pain of unclear etiology. To minimize frustration in patients, parents, coaches, and physicians, it is important to have a clear treatment algorithm for these situations. Patients with neurologic-type symptoms in a nonanatomic distribution should be examined for neural tension signs more proximally, and nerve gliding exercises may be the key to achieving return to sport. Nonspecific wrist pain is common due to ligamentous laxity and muscular immaturity. A neuromuscular training program emphasizing resistive strengthening relieves symptoms in most patients.² A baseline therapy evaluation for grip and pinch strength is followed by a 6-week exercise program while concomitantly protecting the wrist joint and modifying activities. Patients who fail to improve their strength should continue exercises. If their pain is ongoing despite improved strength, MRI evaluation is undertaken. As always, it is important to validate symptoms and tailor treatment to patient goals.

FINGER FRACTURES

Fractures account for two-thirds of hand injuries in the under 16 age group.¹ Skeletally immature patients are vulnerable to unique patterns of injuries, as the weak points in their anatomy are the growth plate and subcapital region of the phalanges. The Seymour fracture is an open fracture of the distal phalanx through the physis that can present similar to a mallet injury, with about half occurring during sports.³ Subcapital fractures of the phalangeal neck appear deceptively benign and therefore often have a delayed presentation. Phalangeal base fractures are most commonly seen in the small finger proximal phalanx, 22 of 34 fingers in one series,⁴ where they carry the eponym “extra octave fracture” due to ulnar dorsal angulation giving the appearance of the small finger trying to reach further on piano keys. Metacarpal fractures can occur in any age group, but are most common in 10- to 19-year-olds involved in basketball, football, or bicycling.⁵

History

Seymour fractures often occur with hyperflexion of the distal interphalangeal (DIP) joint similar to a mallet finger, though many describe a “crushing” component to their mechanism. Bleeding at the time of injury distinguishes an open fracture from a closed nailbed or distal tuft injury. Prior antibiotic treatment should be determined.

Subcapital fractures also occur after crush injuries in sports, with immediate pain and swelling. Bruising may be minimal in this relatively avascular region. Motion, specifically flexion, is limited. With restricted motion, patients can be relatively painless, often leading to delayed presentation.

Phalangeal base fractures occur most commonly with a dorsal ulnar pull on the digit, resulting in immediate pain and deformity. Reduction may or may not have been attempted on the field. Boxer’s fractures occur after direct impact of the dorsal 5th metacarpal head. True boxing is a rare cause, as punching should axially load the 2nd and 3rd metacarpal heads. Patients will notice decreased prominence of the knuckle. If patients complain of repetitive snapping or popping, one should suspect a “boxer’s knuckle,” or sagittal band rupture allowing subluxation of the extensor tendon.

Physical Examination

Swelling, integrity of the nail and nailfold, ecchymosis, and resting position should be noted. A Seymour fracture may have been splinted in extension with the proximal nail resting superficial to the eponychial fold. This can give the subtle appearance of a longer nail. There will be an extension lag at the DIP joint. Dried blood should be noted, as well as any signs of infection in late presenting cases. As with any injury, concomitant injuries should be looked for, with the most common being additional finger fractures.³

Phalangeal neck fractures will present with a swollen proximal interphalangeal (PIP) or DIP joint, depending on whether the fracture is in the proximal or middle phalanx, respectively. Tenderness to palpation is important, as this indicates the fracture is mobile and more amenable to percutaneous reduction. Flexion is limited due to pain and angulation. Coronal plane angulation is examined in extension, while malrotation is best assessed for with flexed digits to demonstrate under- or overlapping. In patients unable to flex due to pain, passively extending the wrist will utilize the tenodesis effect to flex the digits ([Fig. 135.1](#)). This

Abstract

Wrist and hand injuries are common in adolescent athletes. Certain conditions are unique to this age group, while others are found in adults but have distinct treatment considerations in skeletally immature patients. Seymour fractures are nailbed lacerations with underlying open distal phalanx physeal fractures that are biomechanically similar to mallet injuries but managed differently. Subcapital phalangeal neck fractures are rare in adults but common and problematic in children if not promptly diagnosed and stabilized. Salter Harris II fractures are common in both the base of the small finger proximal phalanx and 5th metacarpal neck, and both can usually be managed with reduction as needed, combined with buddy strapping.

Jamming injuries of the finger may result in volar plate, collateral ligament, or central slip injury. The former two injuries rarely have sequelae and can be allowed early range of motion. Central slip injuries can be more difficult to diagnose and may result in boutonnière deformity if not immobilized properly.

Wrist injuries are common in athletes of all ages. Distal radial physeal stress reaction is unique to adolescents, as their open growth plates are vulnerable with repetitive loading activities. Scaphoid fractures, once a rare injury in children, are increasingly common. A growing body of literature exists to guide treatment in the adolescent age group. Finally, the frustration of nonspecific wrist pain can be mitigated by a structured treatment algorithm.

Keywords

wrist pain
phalangeal fracture
hand injury
jammed finger
scaphoid
distal radial physis



Fig. 135.1 Wrist extension utilizing tenodesis effect to flex digits, revealing digit scissoring, or overlapping, from malrotation deformity.

can be done while palpating the scaphoid tubercle, as all digit tips should point toward the tubercle in flexion.

Phalangeal base fractures will have obvious hyperextension and ulnar angulation around the metacarpophalangeal (MCP) joint. Making a full fist may be still possible due to composite flexion. Boxer's fractures demonstrate a depressed knuckle. Rarely the apex dorsal angulation will cause skin tenting, threatening open injury due to thin skin and tendon covering.⁶ Malrotation is uncommon but crucial to assess for. Contralateral comparison is important, as there are varying degrees of baseline digit angulation.

Imaging

Anteroposterior (AP), lateral, and oblique radiographs of the digit should be obtained for all finger injuries, as there will be less digital overlap on the lateral and oblique views compared with hand radiographs, which are more appropriate for metacarpal fractures. Resting position and soft tissue swelling are important to note in addition to the fracture. Seymour fractures will show flexion through the distal phalangeal physis (Fig. 135.2). If held in extension, however, the injury can be subtle with widening of the physis alone.

Subcapital fractures are often missed on the AP view, as hyperextension causes bony overlap. If not familiar with skeletally immature radiographs, the interpreter may mistake the transverse fracture line for a physis. Physes should be proximal in the phalanges instead of the distal neck region. One should check for intra-articular extension, as intercondylar fractures tend to be more unstable and will often flex and shorten, even when initially nondisplaced. The lateral view will typically demonstrate extension of the distal fragment. Malrotation can be a subtle "double bump" of the condyles of the phalangeal head on a true lateral where adjacent condyles are collinear (Fig. 135.3A).

Phalangeal base and metacarpal fractures are best assessed on oblique views of the finger and hand, respectively, as other views minimize apparent angulation. These fractures likely involve



Fig. 135.2 Lateral and oblique radiographs demonstrating an open fracture through the physis of the distal phalanx, or Seymour fracture, in the typical posture of flexion.

the physis in a Salter Harris II pattern. Angulation is typically dorso-ulnar in the small finger proximal phalanx (Fig. 135.4A) and palmar-radial in the metacarpal neck.

Decision-Making Principles

Hand fractures will often present late, as athletes are able to continue playing, and fear of restrictions may lead patients to downplay their injury. Radiographs should be obtained with any history of trauma and immediate swelling. Juxta-articular soft tissue injuries and unstable finger fractures have acute presentations too similar to distinguish accurately by clinical exam alone. Examination under digital block is poorly tolerated in the adolescent patient.

Both Seymour and subcapital fractures are inherently unstable; therefore operative stabilization is indicated in all but absolutely nondisplaced fractures. Phalangeal base and metacarpal neck fractures are close to both the physis and the 5th MCP joint with multiplanar motion. This allows for remodeling as well as better tolerance of angulation.⁷

Treatment Options

Nailbed injuries with underlying closed distal tuft fractures are stabilized by the nail itself and managed nonoperatively. Due to instability, Seymour fractures are operative injuries involving a combination of antibiotics, nailbed repair, fracture reduction, and stabilization. One study recommended oral antibiotics be given immediately and continued 4 days postoperatively in any case treated over 6 hours after presentation.³ Nailbed repair was traditionally employed with a fine absorbable suture such as 6.0 chromic; however, a randomized control trial (RCT) demonstrated faster repair time and similar results using 2-octylcyanoacrylate (Dermabond; Ethicon Inc, Somerville, NJ).⁸ Fracture stabilization produces a more favorable environment for soft tissue healing and theoretically minimizes infection in Seymour fractures. Splinting alone risks late displacement; therefore pinning is often employed.

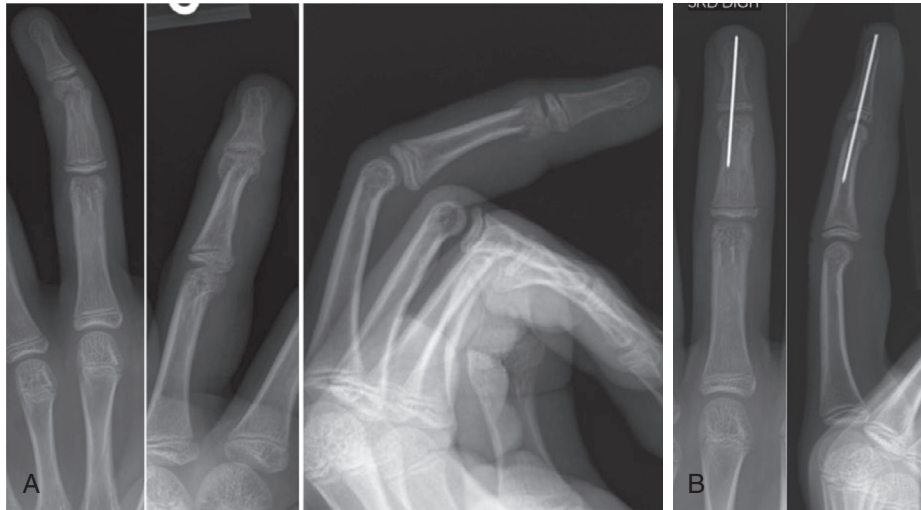


Fig. 135.3 Radiographs of a 12-year-old patient's finger (A) demonstrating a middle phalangeal neck fracture in ulnar deviation, extension, and malrotation. The degree of malrotation is such that the lateral view of the digit shows both the condyles, as it is an anteroposterior (AP) of the fracture fragment. AP and lateral after closed reduction and pinning (B) show restoration of anatomic alignment and stabilization with a buried intrafocal pin. Note the lateral demonstrating overlap of the condyles of both the proximal and middle phalanges, confirming correction of malrotation, as this is difficult to assess clinically once an interphalangeal joint is pinned in extension.



Fig. 135.4 Anteroposterior and lateral radiographs in a 16-year-old show a small finger proximal phalangeal base Salter Harris II fracture in the typical position of extension and ulnar deviation (A). Closed reduction under digital nerve block was followed by buddy strapping and post reduction radiographs (B) showing improved alignment.

Nonoperative management is rarely indicated for subcapital fractures. Nondisplaced fractures can be immobilized in extension with weekly radiographs for the first 3 weeks to ensure maintained alignment. While splinting is not ideal in this often-noncompliant age group, it allows for better lateral radiographs. Digital overlap will obscure a good lateral if casted. Displaced fractures are ideally close reduced and percutaneously pinned (see Fig. 135.3B). Phalangeal neck fractures over 3 weeks from injury may be difficult to reduce closed. Percutaneous osteoclasis can be attempted; however, open reduction should be undertaken with caution, as the tenuous blood supply of the mostly articular

distal fragment through the collaterals can be disrupted. Internal fixation with headless screws can be considered to allow early postoperative motion, as the PIP joint is prone to stiffness.

Phalangeal base fractures are mostly amenable to nonoperative management. Within the first 2 weeks, fracture site mobility should allow closed reduction with ligamentotaxis with or without digital nerve block (see Fig. 135.4B). The broad fracture surface across the physis and metaphysis lends it to be relatively stable. Acutely irreducible fracture has been reported with entrapped tendon and is a rare indication for open reduction. If operative reduction is undertaken, Kirschner wire (K-wire) stabilization



Fig. 135.5 Radiographs of the hand (A) demonstrate a Salter-Harris II 5th metacarpal neck fracture, known as a boxer's fracture. Maximum angulation is noted on the oblique view in the typical palmar radial direction. Oblique view 2 weeks postinjury (B) demonstrates maintained alignment and callus formation in a patient managed with buddy strapping.

is performed by entering the head of the proximal phalanx and driving the pin across the flexed MCP joint, leaving the PIP free.⁴ In late presenting injuries that cannot be close reduced, it is prudent to allow healing and consider later corrective osteotomy. This allows remodeling and minimizes the risk of iatrogenic damage to the physis.

Boxer's fractures, though relatively unstable, are also amenable to nonoperative treatment in most cases (Fig. 135.5). If reduction is performed, the ring and small fingers are traditionally casted in the intrinsic plus position (MCP flexion and PIP/DIP extension) for 4 weeks. Casting in MCP joint extension up to the PIP joints has been shown to be well tolerated with better range of motion (ROM) and grip strength without any difference in fracture angulation compared with traditional intrinsic plus positioning in a RCT.⁹ Operative indications include open injuries, multiple fractures, malrotation, shortening causing pseudoclauing, and strong patient/parent preference. Multiple percutaneous pin configuration options have been described. Internal fixation options exist, such as a condylar blade plate, but involve crossing the physis in adolescents and may become stress risers for additional fractures with subsequent trauma.

Postoperative Management

Patients with K-wires are splinted or casted for protection of the fracture and hardware. Pins left percutaneously out of the skin are removed in the clinic around 4 weeks postinjury (later in cases where osteoclasts was required), after checking radiographs to confirm provisional healing. Buried pins that do not prevent PIP motion, such as those crossing only the DIP joint, can be left longer and removed after injection of local anesthetic. Removable splints allowing early ROM are employed until fractures are fully consolidated around 6 weeks postinjury. Protocols are modified

for athletes wishing to return to sport early on the basis of the individual and the sport's rules on playing with braces.

Results

Seymour fractures generally do well, with one series showing 23 of 24 patients regaining full motion with good satisfaction and no cases of infection.³ Phalangeal neck fractures treated with percutaneous pinning were recently shown to have 92% good to excellent results.¹⁵ No patients had open reduction, as percutaneous osteoclasts was successfully performed up to 36 days after injury.

Phalangeal base and boxer's fractures have similar outcomes. In a series of 34 phalangeal base fractures, all but two severely displaced fractures healed without complications or residual deformity with full ROM.⁴ For boxer's fractures, a prospective study of fractures angulated up to 75 degrees treated with Coban self-adherent wrap (3M, St. Paul, MN) and buddy tape showed all patients to be satisfied and functionally unimpaired with no loss of grip strength.¹⁶ A RCT demonstrated final fracture alignment to be no different with this soft wrap method compared with reduction and casting.¹⁷

Complications

One series of Seymour fractures reported complications in half of patients, though the majority were minor, such as nail and subtle distal phalanx growth disturbances. Secondary fracture displacement can occur if stabilization is not performed.³

Subcapital fracture nonunion is reported, usually in thumbs managed nonoperatively, while malunion is more common in the digits.¹⁸

Phalangeal base fractures treated with open reduction or late osteotomy carry a risk of stiffness from tendon adhesions



Author's Preferred Technique

Seymour fractures, when possible, have antibiotics and tetanus prophylaxis administered acutely. Under finger tourniquet control, the nail plate is removed, as it blocks reduction and cleansing of the open fracture. The Betadine-cleansed nail or a piece of Xeroform dressing is placed under the eponychial fold to splint it open, and the fracture is reduced with a combination of extension and volar directed translational pressure. It is important to hold this reduction, as the fracture is unstable, and multiple reductions can damage the physis. A K-wire, usually 0.045 in adolescents, is inserted into the digit tip and driven across the physis and distal interphalangeal (DIP) joint to rest in the middle phalanx. The K-wire is cut and left buried. Dermabond is used to repair the nailbed, secure the nail plate, and cover the pin site, except in cases with concern for infection, as drainage should be allowed. I do not routinely prescribe antibiotics postoperatively, and instead reserve it for cases that appear contaminated.

Displaced, angulated, or rotated proximal phalangeal neck fractures are managed with closed reduction and intrafocal pinning when possible. I utilize a technique of proximal interphalangeal flexion and rotation to obtain reduction, followed by K-wire insertion through the head of the proximal phalanx. The wire is driven out through the base of the proximal phalanx, holding the MCP joint in full flexion and pulled proximally until it rests just subchondral in the proximal phalanx head. The PIP joint is extended to allow the wire to be driven into the middle phalanx for additional fixation.¹⁰ One 0.045 K-wire will typically hold the digit in the intrinsic plus position and incur sufficient stability; however, this can be supplemented with an additional wire for rotational control as needed. Pins are left percutaneous, and a protective overlying cast is placed.

Middle phalangeal neck fractures are managed similarly,¹¹ with the difference that slight angulation and flexion deficits are better tolerated near the DIP joint. For subacute fractures that are still tender, I will perform dorsolateral percutaneous osteoclasts with a K-wire. I do not risk avascular necrosis (AVN) with open

reduction. Some sagittal remodeling may occur^{12,13} through the growth plate under the cartilage cap. For limitation of flexion alone, volar bone spike can be removed to clear the subcondylar fossa and tends to improve motion.¹² For malrotation or angulation, corrective osteotomy can be performed in a safer region more proximally in the shaft.

Phalangeal base fractures are best managed with closed reduction under digital nerve block. Entry into the ulnar aspect of the palmodigital crease allows lidocaine infiltration proximally about both radial and ulnar digital nerves, as well as dorsally for a complete block. This is better tolerated than a needlestick into the sensitive palm. For the small finger, a fulcrum such as a thin pen is placed into the 4th webspace. The small finger is brought over this fulcrum into hyperflexion and maximum radial deviation. A palpable reduction is typically appreciated. Postreduction radiographs are obtained, though clinical appearance and ROM determine acceptable alignment.

Boxer's fractures are managed nonoperatively with the exceptions noted previously. Closed reduction is with the Jahss maneuver, flexing the MCP joint and extending the PIP joint to push the metacarpal head dorsal while relaxing the intrinsic muscles. When no reduction is desired, my preference is for buddy taping as soon as comfort allows. Some patients will request higher degrees of immobilization, in which case various prefabricated boxer's fracture braces are commercially available and stocked in the clinic. Athletes can return to sports as soon as pain and mobility allow,¹⁴ often within 2 weeks once the fracture has stabilized with provisional callus. Operative management is with closed or percutaneous reduction when possible. With a Kapandji technique, a K-wire inserted into the fracture site dorsolaterally can lever on the intramedullary canal of distal fragment as needed. Stabilization is performed with 2 K-wires, typically 0.045 or 0.054, inserted into the dorsal metacarpal head from ulnar and radial and driven to the base of the metacarpal, left percutaneous.

requiring tenolysis. Malunion in hyperextension for phalangeal base fractures or hyperflexion for boxer's fractures may cause a pseudo-claw deformity from extensor shortening.⁴

Future Considerations

Studies of Seymour fractures and all nailbed and open distal phalanx fractures are needed to guide appropriate antibiotic use. Subcapital fractures are increasingly studied; however, specific factors associated with delayed presentation remain unknown. For phalangeal base and metacarpal fractures that remain angulated, it is difficult to rely on remodeling when there is limited growth potential remaining, as is often the case in adolescent athletes. Further study into the rate of angular correction in various planes for hand injuries would help guide treatment.

JAMMED FINGER

Soft tissue injuries account for one-third of hand trauma in patients under 16 years,¹ and the PIP is the most common location of digital injury in sports. Common injuries affecting the PIP joint include contusion or rupture of the volar plate, radial and ulnar collateral ligaments (RCL and UCL), and central slip. The volar plate injury is the second most common injury in softball, second to mallet fractures.¹⁹

History

Patients may report a specific hyperextension mechanism or only remember the finger was "jammed," typically by a ball

impacting the fingertip. A history of joint dislocation or deformity at the time of injury should be elicited along with any appreciation of a "pop" at the time. Predictable injury patterns occur depending on the direction of dislocation. Dorsal dislocation injures the volar plate, lateral dislocation affects the collaterals, and volar dislocation is associated with central slip tear. Central slip injury is also caused by direct dorsal impact or forced PIP flexion.

The chronicity of the injury and type and length of immobilization should be clarified. In addition, patients should be asked about pain, limitation of function/ROM, or both. The differential diagnosis for a flexed digit includes pulley rupture, damage to extensor tendon, fracture, and dislocation.²⁰ Less common is the extended digit, which may be due to flexor rupture or a volar dislocation. Rapid forced flexion or volar dislocation will result in central slip disruption with or without a dorsal middle phalanx fracture.

Physical Examination

Bruising and swelling can persist long after tenderness has resolved. Resting position and ROM should be checked. Normal PIP ROM is from 0 to 110 degrees. Passive PIP extension may cause discomfort or demonstrate lack of a firm endpoint with volar plate injury. Contracture of the injured volar plate leads to pseudo-boutonnière deformity, or PIP joint flexion contracture without DIP deformity.

Boutonnière deformity will be present if the central slip and triangular ligament both tear and allow the lateral bands

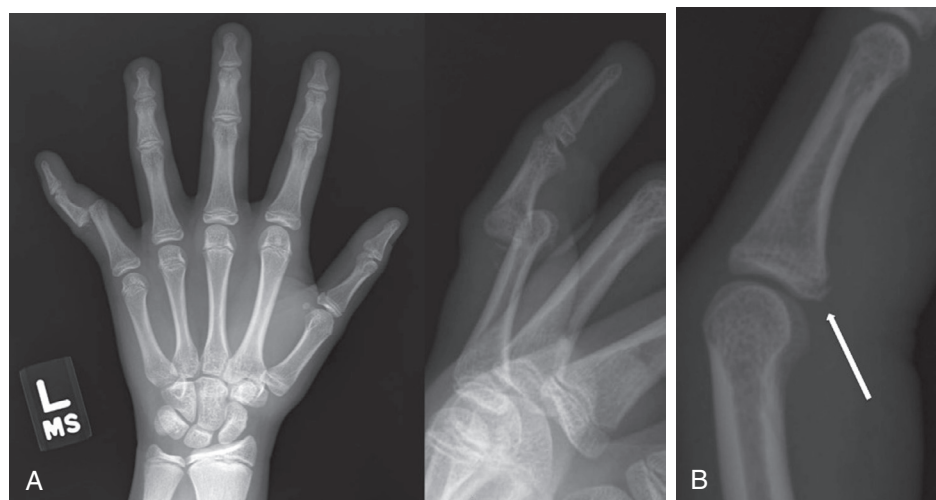


Fig. 135.6 Posterolateral dislocation of the small finger proximal interphalangeal joint is seen on anteroposterior and lateral radiographs (A). A close-up of the postreduction lateral radiograph (B) demonstrates a minimally displaced Salter Harris II fracture of the volar middle phalanx base (*arrow*), which represents a stable avulsion fracture of the volar plate.

to subluxate volar to the PIP joint. As this is not always present acutely, a high degree of suspicion for central slip injury is necessary and should be evaluated with the Elson test, with digital block for pain as needed. When the PIP joint is flexed, the DIP is normally unable to be actively extended due to slack in the lateral bands. DIP hyperextension via lateral bands during resisted extension of the flexed PIP indicates central slip injury.

Varus and valgus stress of the PIP joint in full extension and 30 degrees of flexion tests the accessory and proper collaterals, respectively. While the index finger RCL is most commonly injured, it is more common to have a combination RCL and UCL injury.

Imaging

Finger radiographs are examined for joint congruency, resting position, and any fractures. With volar plate injuries, an avulsion fracture from the middle phalangeal base may be seen, present in 38% of injuries in one study.¹⁹ In skeletally immature patients, this may be a Salter-Harris III or IV pattern involving the physis with an epiphyseal component with or without a metaphyseal component (Fig. 135.6). Dorsal fracture may be present with volar dislocation or central slip avulsion. Collateral ligament injury may be accompanied by avulsion fracture in the subcondylar fossa of the proximal phalanx. Stress radiographs can reveal joint widening but are usually not tolerated acutely and are unnecessary with chronic injury, as physical exam is diagnostic. Advanced imaging with high resolution MRI can demonstrate further detail but is rarely necessary.

Decision-Making Principles

Painless full ROM with a stable joint is the goal. This is achieved with brief periods of immobilization to allow soft tissue healing followed by mobilization to prevent contracture. Unstable joints must be stabilized prior to allowing ROM.

Treatment Options

Dorsal or lateral dislocations are generally able to be close reduced. Volar dislocation may have soft tissue interposition or a rotatory component with the lateral condyle buttonholed between the collateral ligament and lateral band. MCP and PIP flexion may allow the digit to be rotated back into place; otherwise open reduction is indicated. Postreduction, the digit should be examined for stability, rigidly immobilized, and radiographed.

Most soft tissue injuries and small bony avulsions can be mobilized early and buddy strapped. Sports can be continued with protection, though use of the rigid splint may compromise glove fit and grip strength. In rare cases, primary repair of soft tissues such as collateral ligaments is indicated if interposed into the joint and requiring open removal, though stiffness is common following this repair.²⁰

The exception to early motion is acute central slip injury, which is treated with 6 weeks of full time extension splinting of the PIP joint followed by nighttime splinting for an additional 6 weeks to prevent late boutonnière deformity. DIP motion is encouraged while splinting to draw lateral bands into the dorsal position.²¹ Open injuries and large dorsal fractures are operative.



Author's Preferred Technique

Acute nonoperative management is as above. Patients that present subacutely with proximal interphalangeal (PIP) stiffness undergo static progressive splinting. To achieve extension, the splint includes a point of counterforce on the palmar aspect of the metacarpal head to prevent MCP joint hyperextension while working on PIP motion.

An associated fracture of the middle phalanx involving more than 30% of the articular surface with associated PIP instability is an indication for operative stabilization. Similar to adults, percutaneous pinning, dynamic traction,²² or internal fixation with mini-fragment screws can be employed. Hemi hamate arthroplasty for volar fractures and suture anchor repair of central slip dorsal avulsion are also considered.^{23,24}

Postoperative Management

Motion exercises are started 2 weeks postoperatively for most injuries, or until percutaneous pins are removed. Early return to play (RTP) with the PIP joint protected in a rigid cast can be considered for certain sports; however, if PIP extension will hinder function or splinting is not allowed, the athlete will be restricted for a minimum of 6 weeks.

Results

Children with PIP hyperextension injuries have better outcomes than adults. An RCT showed buddy strapping for a week to produce better resolution of edema and pain compared with a group treated with a week of Alumafoam extension blocking orthosis.²⁵ Full motion was regained in 65% at 1 week and 91% at 2 weeks.

Collateral ligament injuries of the PIP joint treated with buddy taping for an average of 3 weeks demonstrate 77% recovery of grip strength and 84% recovery of ROM at 6 months. Increased disability is associated with delayed treatment, female gender, and radial digit injury at 3 months.²⁶

Complications

Untreated injuries can lead to pain, deformity, instability, and degenerative joint disease.²⁰ Boutonnière deformity can develop within 2 to 3 weeks after central slip tear. PIP flexion contractures not responding to progressive splinting may need open release.

DISTAL RADIAL PHYSEAL STRESS REACTION

More than half of gymnasts experience wrist pain, usually with more than 6 months of duration.²⁷ Ninety percent of those with wrist pain still have it a year later. The differential includes both bony conditions (distal radial physeal stress, impaction syndromes, scaphoid stress fracture, carpal chondromalacia, avascular necrosis [AVN] of the capitate) and soft tissue abnormalities (dorsal impingement, triangular fibrocartilage complex [TFCC] tears, ganglia, carpal, or distal radial ulnar joint [DRUJ] instability).

The skeletally immature population tends to have negative ulnar variance that becomes more negative with maturation, as the distal ulna physis closes before that of the distal radius. The more ulnar negative the wrist, however, the more an axial load is borne by the distal radial physis. Gymnasts, by contrast, are relatively ulnar positive, and become more so with increasing years of practice and higher body weight.²⁸ While this theoretically protects the distal radial physis, pathologic dynamic loading of the wrist in ligamentously lax gymnasts can still occur. The mechanism of “gymnast’s wrist,” or physeal stress injury, is likely a combination of vascular insufficiency and mechanical growth inhibition in accordance with the Hueter-Volkmann law.² In addition, ulnar positive wrists are more likely to demonstrate TFCC pathology and dynamic ulnocarpal impaction.

History

The history is the key to diagnosis. Patients often complain of aching dorsocentral wrist pain during or after loaded extension activities. Specific gymnastic risk factors include soft mats that

lend to increased wrist dorsiflexion, twisting vault routines that produce dorsiflexion with ulnar deviation, and beam activities with locked forearms that transmit rotational moments to the wrist.²⁷ The physis is most vulnerable during rapid growth, occurring on average around years 10 to 14.

Physical Examination

Distal radial physeal tenderness is often present, especially dorsally. The push-up test, loading the wrist in extension as when pushing up out of a chair, may reproduce radial or ulnar pain from distal radial physeal loading or dynamic ulnar carpal impaction, respectively. There may be a subtle loss of wrist extension and grip strength. These findings are also present with occult ganglia and dorsal impingement; therefore location of tenderness is a key feature. Long-standing cases with physeal growth disturbance may have a prominent ulnar styloid from relative overgrowth. The DRUJ should be examined for stability and compared with the contralateral side, as ulnar overgrowth may lead to an incongruent articulation with the sigmoid notch.

Imaging

Stress-related distal radial physeal changes are present on radiographs in 10% of gymnasts.²⁸ Findings include widening of the physis, irregular borders, radial-sided cystic changes, a beaked epiphysis, and haziness within the physis. Ulnar variance should be measured on an AP taken in neutral rotation. The Hafner method of measuring ulnar variance involves comparing the most proximal and distal extents of the distal radial and ulnar physes (Fig. 135.7).²⁹ The distal extent of epiphyseal ossification is not as reliable as the thickness of the unossified portion of the distal radius and ulnar are not known. Late-stage deformity

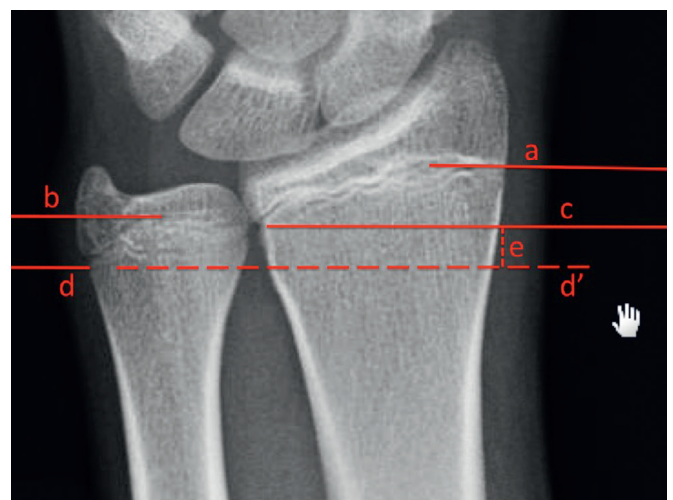


Fig. 135.7 Ulnar variance in a skeletally immature wrist is best assessed on an anteroposterior taken in neutral rotation, with the shoulder abducted 45 degrees and the elbow flexed to 90 degrees. The vertical distances between lines tangential most distal (a to b) and proximal (c to d) portions of the distal radial and ulnar metaphyses are measured. The line from the most proximal ulnar metaphysis portion (d) has been extended (d') to demonstrate how to measure the variance (e).

can involve a Madelung-like deformity with a volar and ulnar directed distal radial articular surface.

MRI has not been shown to be a useful prognostic tool.³⁰ All asymptomatic college gymnasts show MRI changes in the wrist ligaments, tendons, and cartilage.³¹ Eighty percent also show bony changes and cysts/fluid collections, despite remaining asymptomatic for several years.

Decision-Making Principles

Three stages have been described to guide treatment, with the mainstay being activity restriction. Stage 1 is a clinical diagnosis without radiographic changes, and management is gradual return to sport after symptoms cease with complete rest. Stage 2 is when radiographic changes are present, and return to sport averages 2 to 4 months. Stage 3 demonstrates characteristic radiographic changes along with positive ulnar variance. It is unclear how reversible the injury is once growth disturbance has occurred.

Treatment Options

While prevention of repetitive wrist loading would be the best treatment for the general health of the wrist, this is often not an option for elite athletes. An initial cessation from weight bearing on the wrist is recommended until symptoms resolve. Despite a lack of evidence, bracing is often employed upon return to sport, with restrictions implemented again if symptoms recur. Recommendations from Waters and Bae include cyclic progressive training cycles, alternating loading (pushing activities 1 day, pulling the next), bracing such as tiger/lion paws, and a regimen of 6 weeks of rest followed by 6 weeks of closed chain exercises prior to resuming open chain activities.² They emphasize that restrictions should be guided by soreness rather than absolute time periods.

Radiographic follow-up of the wrist is prudent to assess for ongoing growth disturbance. Progression of ulnar positive variance is an indication for epiphysodesis of the distal ulna with ulnar shortening as needed.



Author's Preferred Technique

Due to high rates of noncompliance with elite athletes, specifically gymnasts, I initially institute a period of short arm casting for 4 weeks to ensure rest. This is followed by continued restriction of loaded extension activities while initiating ROM and grip strengthening exercises. While radiographs rarely show resolution of physeal changes, follow-up radiographs are obtained around 8 weeks to check for progression of any positive findings. In the absence of progressive changes, the athlete is gradually returned to sport with brace protection when possible and instructions to cease loading if symptoms recur.

Indications for distal ulnar epiphysodesis include an open physis with progressively positive ulnar variance or symptomatic ulnocarpal impingement, and in the latter case this is combined with ulnar shortening osteotomy. Epiphysodesis is performed with a direct ulnar approach and drilling the physis in all directions. Systematic spherical curettage is then performed until the soft feel of the physis has given way to the rough scraping of cancellous bone. The distal ulnar epiphysis is notorious for incomplete epiphysodesis, and therefore this is performed quite meticulously.^{32,33} In the case of suspected TFCC pathology, wrist arthroscopy is concomitantly performed.

Postoperative Management

Epiphysodesis is followed by 2 weeks of splint protection and early mobilization with protected lifting for an additional 4 weeks. Ulnar shortening osteotomy is followed by lifting restrictions until radiographic healing—typically 8 weeks.

Results

No studies have demonstrated specific outcomes of treatment; however, symptomatic relief is reported with activity restriction.

Complications

Growth arrest has been reported; however, the incidence and preventability is unknown.

FUTURE CONSIDERATIONS

Prospective studies to quantify the “dose” of loading that leads to this injury would be helpful in guiding training.²⁷ As with pitch counts in baseball, injury may be prevented by limiting the quantity of high impact wrist activities that gymnasts undertake in a particular competition or training season.

SCAPHOID FRACTURE

Scaphoid fractures in children and adolescents are increasing owing to increased sports participation. While radiographs are reported to miss the fracture in 2% to 6% of adult cases, misdiagnosis is more common in children due to its relative rarity in children, accounting for only 3% of hand and wrist fracture, and difficulty interpreting skeletally immature radiographs.³⁴

History

The most common mechanism is a fall onto the outstretched radially deviated and pronated hand. There is immediate pain and swelling, though many continue their sport. Radial-sided pain persists, and patients may complain of a weak wrist with limited motion. Timing of injury is relevant, as one-third of pediatric scaphoid fractures present late with chronic nonunion.³⁵ Reasons for delayed presentation including moderate symptoms and fear of losing one's position on the team.³⁴

Physical Examination

Classic tenderness is in the anatomic snuffbox between the abductor pollicis longus and the extensor pollicis brevis, corresponding to the scaphoid waist. Volar pain (over the distal tubercle), pain with radial deviation, and pain with active wrist motion are associated with higher likelihood of fracture in children.³⁶ Non-acute injuries should have assessment of ROM and carpal stability.

Imaging

In addition to the standard three wrist views, specialized clenched fist and ulnar deviation views will dorsally rotate the scaphoid in order to orient the fracture line more parallel to the radiograph beam.³⁷ Seventy-one percent of fractures in adolescents are of the waist.³⁵ If the distal pole is flexed, it will demonstrate the “cortical ring sign,” which is the tubercle seen on fos. The

proximal fragment should be examined for sclerosis, especially in late presenting injuries and proximal pole fractures, as this may indicate AVN. The significance of this finding as it correlates to clinical healing is unclear. In addition, one should be aware of the rare bipartite scaphoid, as this can mimic scaphoid fracture, though some believe this to represent asymptomatic pseudarthrosis of the scaphoid.³⁴

The scapholunate interval should also be examined for widening. While 2 mm is typical in adults, skeletally immature patients have a wider interval averaging 9 mm at age 7. Contralateral comparison should be made, though carpal ossification may not be symmetric.³⁴ Lateral radiographs should be examined for carpal alignment. Lines along the axes of the scaphoid and capitate, as well as the perpendicular to the lunate, are used to measure the scapholunate angle, normally 30 to 60 degrees.³⁷ Both displaced scaphoid fractures and dorsal intercalated segment instability secondary to scapholunate ligament injury will have an increased scapholunate angle from flexion of the distal scaphoid pole and extension of the lunate.

MRI is cost effective for suspected fracture in the face of negative radiographs. It will reveal low signal fracture line surrounded by marrow edema. It can also demonstrate other causes of pain, such as additional carpal fractures, bone contusions, or soft tissue injuries.³⁷ One study of early MRI within a few days of injury for all suspected scaphoid fractures in children demonstrated 22% to have scaphoid fractures. Other injuries included other carpal fractures in 19%, distal radius fracture in 7%, and no distinct injury in 23%.³⁸ MRI is the current gold standard for advanced imaging, though ultrasound and CT can also be utilized.

Decision-Making Principles

Considerations for treatment include fracture orientation, location, displacement, angulation, chronicity, proximal pole radiographic appearance, and patient factors such as tolerance of immobilization.

Treatment Options

As initial radiographs may not clearly demonstrate a fracture line, it is recommended that all clinically suspected scaphoid fractures in children be immobilized with repeat radiographs in 2 weeks, where 30% of radiographically occult fractures become evident.³⁶ Early MRI can be considered for earlier definitive diagnosis. Proximal pole and displaced, angulated, or unstable waist fractures are managed operatively similar to adults.

Nondisplaced waist fractures are amenable to either nonoperative or operative management. Comparison of long and short arm casting in a RCT demonstrated similar union rates but faster healing with the former.³⁹ A higher extent of union on CT was seen at 10 weeks post-casting, with short arm casts that excluded the thumb.³⁹ Subacute fractures, up to 6 months postinjury, have also been shown to have 96% union rates with delayed casting.⁴⁰ RTP is safe 4 weeks after injury in a playing cast for nonoperative fractures.¹⁴ Operative management is with headless cannulated compression screw often placed percutaneously from a volar or dorsal approach, which can be arthroscopically assisted (Fig. 135.8).



Author's Preferred Technique

Short arm casting excluding the thumb is applied for clinically suspected scaphoid fracture regardless of radiographic confirmation, as even in the absence of fracture, this assists in recovery of the contusion and/or soft tissue injury. Occult injuries have repeat radiographs performed out of cast in 2 weeks if tenderness persists. A low threshold for MRI is employed for patients desiring return to sport out of cast.

Nonoperative treatment consists of casting for 6 to 12 weeks, depending on when union is apparent on radiographs. A limited-cut CT is often performed to confirm bridging bone around 8 weeks postinjury if radiographs are not definitive. This regimen is applied to all nondisplaced distal pole and incomplete waist fractures, along with select nondisplaced waist fractures. All displaced waist and proximal pole fractures along with a subset of nondisplaced waist fractures are managed with headless cannulated compression screws. My preference is to perform percutaneous reduction when possible from either a dorsal or volar approach, the latter with the technique detailed nicely by Zlotlow.⁴¹ Fractures requiring open reduction are approached from the dorsal side for reduction and fixation. Chronic injuries and nonunions are supplemented with bone graft placed through the drilled screw tract prior to hardware application. Pedicled bone grafting is reserved for cases with high suspicion of avascular necrosis based on MRI appearance of the proximal pole or intraoperative assessment of bone quality. My preference is for a 1 to 2 intercompartmental suprarotational artery graft fixed with K-wires.

Postoperative Management

A short arm volar splint is placed for 2 weeks postoperatively, followed by prefabricated removable brace application to protect the fracture during healing while allowing removal for ROM exercises. RTP varies from 2 weeks postoperatively in a playing cast for noncontact sports to 8 to 12 weeks or until CT-proven union for other sports such as wrestling and gymnastics.¹⁴

Results

A retrospective review of 351 fractures in patient under 18 years found union rates of 90% and 96.5% in nonoperative and operative fractures, respectively.³⁵ Children that go on to union have excellent functional outcomes regardless of operative or nonoperative management at long-term follow-up.⁴² A meta-analysis of RCTs comparing operative and nonoperative treatment in adults demonstrated operative treatment to produce better functional outcomes, patient satisfaction, grip strength, time to union, and time off work, despite higher complication rates.⁴³ There was no difference in pain, ROM, cost, or malunion/nonunion rates.

Lower union rates are found with casting alone for chronic, displaced, and proximal fractures in children.³⁵ Late presentation and osteonecrosis are each independent predictors of worse outcome in children.⁴² One series of 13 nonunions in children, 75% from sports injuries, demonstrated all to unite after operative stabilization, usually with concomitant bone grafting.⁴⁴

Complications

Scaphotrapezial arthritis occurs more commonly with operatively managed fractures in adults,⁴³ though it is unclear if this is also true of adolescents. There is a small incidence of nonunion, malunion, and proximal pole AVN with any treatment method.

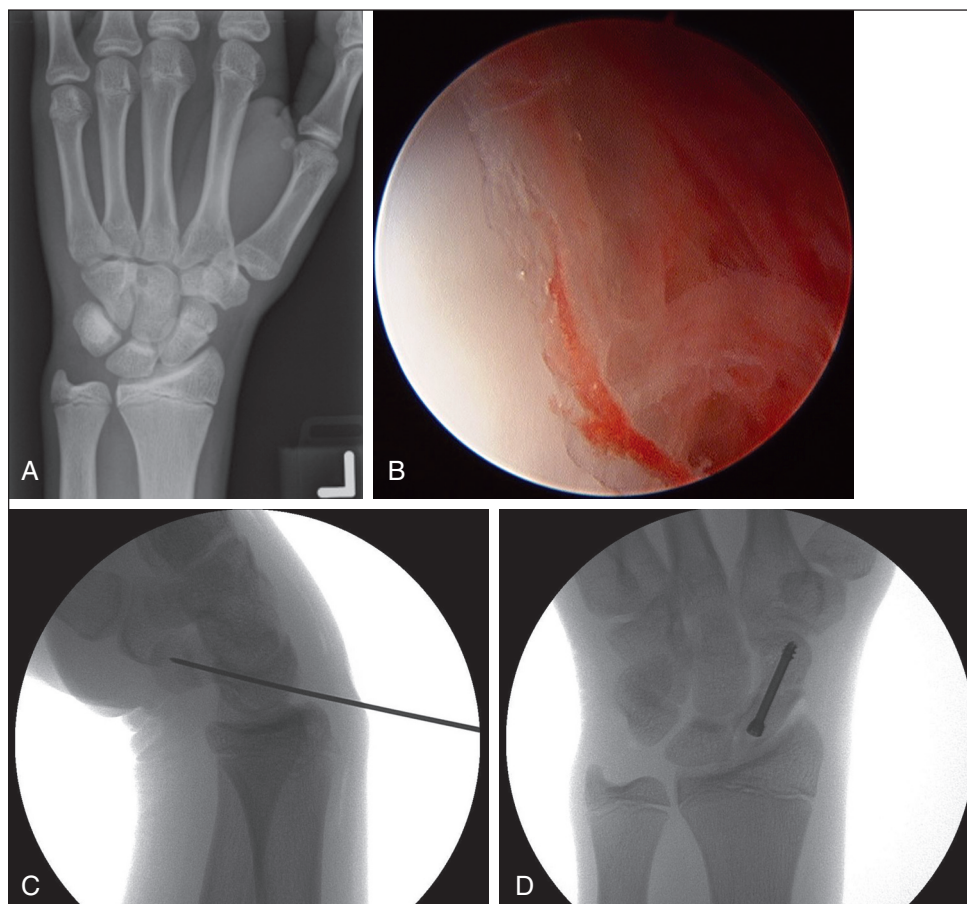


Fig. 135.8 A minimally displaced scaphoid fracture in an adolescent basketball player is demonstrated on anteroposterior radiograph (A). Arthroscopic view of the fracture confirms reduction of the fracture (B). Stabilization is performed with dorsal percutaneous guidewire placement (C) followed by cannulated headless compression screw placement (D).

Future Considerations

Various methods of judging fracture stability and proximal pole vascularity on imaging have been studied,⁴⁵ but the clinical significance of this is unclear. Studies to better predict which fractures go on to nonunion and avascular necrosis would help guide treatment.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Carruthers KH, Skie M, Jain M. Jam injuries of the finger: diagnosis and management of injuries to the interphalangeal joints across multiple sports and levels of experience. *Sports Health*. 2016;8(5):469–478.

Level of Evidence:

V

Summary:

A clinical review of the “jammed” finger is presented and covers pathology at both the PIP and DIP joints. Topics include collateral ligament injuries, pulley ruptures, volar plate injuries, central slip injuries, and finger fractures and dislocations.

Citation:

Lin JD, Strauch RJ. Closed soft tissue extensor mechanism injuries (mallet, boutonniere, and sagittal band). *J Hand Surg Am*. 2014;39(5):1005–1011.

Level of Evidence:

V

Summary:

The authors present a review of the following three traumatic extensor tendon problems: Mallet, boutonniere, and sagittal band injuries. Anatomy, presentation, and treatment are reviewed.

Citation:

Gholson JJ, Bae DS, Zurakowski D, et al. Scaphoid fractures in children and adolescents: contemporary injury patterns and factors influencing time to union. *J Bone Joint Surg Am*. 2011;93(13):1210–1219.

Level of Evidence:

III

Summary:

A retrospective analysis of 351 scaphoid fractures in patients under 18 years demonstrated the most common fracture location to be the scaphoid waist. Nonoperative management led to union in 90%. One-third of patients presented late with chronic nonunion.

Citation:

Matzon JL, Cornwall R. A stepwise algorithm for surgical treatment of type II displaced pediatric phalangeal neck fractures. *J Hand Surg Am*. 2014;39(3):467–473.

Level of Evidence:

IV

Summary:

A series of 61 phalangeal neck fractures treated with closed or percutaneous reduction and pinning are presented. Full functional recovery was achieved in 53 patients. Open reduction was not necessary in any patient, even up to 36 days postinjury.

Citation:

Rettig AC. Athletic injuries of the wrist and hand. Part I: traumatic injuries of the wrist. *Am J Sports Med*. 2003;31(6):1038–1048.

Rettig AC. Athletic injuries of the wrist and hand. Part II: overuse injuries of the wrist and traumatic injuries to the hand. *Am J Sports Med*. 2004;32(1):262–273.

Level of Evidence:

V

Summary:

Athletic injuries of the hand and wrist are reviewed in this two-part paper divided into traumatic and overuse conditions. Scaphoid, hamate, lunate, and pisiform fractures are discussed. Ligamentous injuries of the wrist covered include scapholunate and lunotriquetral injuries, along with midcarpal instability. DRUJ and TFCC injuries are also covered. Overuse injuries of the wrist with different tendinopathies are reviewed. Lastly, finger injuries ranging from fractures to soft tissue injuries are detailed.

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